INCH-POUND

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MILITARY SPECIFICATION

BODY ARMOR, COAT, FOR EXPLOSIVE ORDNANCE DISPOSAL (EOD) SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers body armor coats for the Explosive Ordnance Disposal (EOD) system. The coat includes an insert of aramid ballistic filler enclosed in an oxford aramid cloth shell. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administration and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The body armor shall be of one type and in the following sizes as specified (see 6.2).

Sizes
Extra small
Small
Medium
Large
Extra large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8470

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-543	-	Buckle, Tongueless, and Web Strap
MIL-W-4088	-	Webbing, Textile, Woven Nylon
MIL-T-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-W-5664	-	Webbing, Textile, Elastic
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43128	-	Cloth, Plain Weave, Nylon: Water Repellent, OG-106
MIL-C-43842	-	Cloth, Oxford, Aramid
MIL-C-44050	-	Cloth, Ballistic, Aramid
MIL-T-44100	-	Thread, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-2669 - Toggle-Control Line

(Copies of drawings are available from the U.S. Army Natick, Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

(Application for color cards should be addressed to the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Cloth, oxford, aramid. The cloth for the inner and outer shell, and removable groin protector shall be aramid oxford, Olive Green 106, conforming to MIL-C-43842.

3.3.2 Cloth, ballistic, aramid. The cloth for ballistic filler inserts shall be water-repellent aramid ballistic cloth conforming to type II, class 2 of MIL-C-44050.

3.3.3 Thread. The thread for all stitching throughout the coat shall be para-aramid, spun, intermediate, modulus conforming to MIL-T-44100 and shall be Olive Drab S-1, C.A. 66022. Thread size shall be as specified in table II.

3.3.4 Webbing, nylon. The 1-23/32 inch wide nylon webbing used on the right and left shoulder quick-releases shall be type VIII, class 1, Olive Drab 7, abrasion resistant, conforming to MIL-W-4088.

3.3.5 Webbing, textile, nylon. The 1-1/2 inch wide nylon webbing used for constructing the ballistic inserts when butting seams and the 1 inch wide nylon webbing used for the collar and neck quick-release, chest plate holder and front toggle holders shall be type IV, Olive Drab 7, conforming to MIL-W-5038.

3.3.6 Webbing, elastic. The 1-1/2 inch wide cotton elastic webbing for the left shoulder, shoulder shields, elbow shields, and groin protector shall be Olive Drab 7 conforming to type II, class 1 of MIL-W-5664.

3.3.7 Fastener tape. The 1 inch and 2 inch wide synthetic fastener tapes (hook and loop) used on quick-releases, pocket flaps, chest plate straps, collar and groin protector shall be type I, class 2, color green 3421 of MIL-F-21840.

3.3.8 Buckle. The adjustable buckle used for securing the chest plate shall be quick-release, non-slip, conforming to type V, class 3 of MIL-B-543.

3.3.9 Toggle. The toggle for quick release straps shall conform to Natick drawing 11-1-2669.

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3.3.10 Size, identification, and instruction label. The body armor shall have a combination size, identification, and instruction label conforming to type VI, class 14 of DDD-L-20 with the following exceptions and detailed requirements.

a. The size of characters for the instruction label, except capital letters, shall be 1/16 inch minimum. All capital letters shall be 3/32 inch minimum.

b. The label material shall be nylon conforming to MIL-C-43128, except it shall not be water-repellent treated. The label material shall be dyed Medium Green 614 matching the standard sample (see 6.4).

c. The required permanent coating shall not apply.

d. The combination label shall be printed with the instruction label first and the size and identification label placed below the instruction label. The size and identification label shall be combined with the size label contents placed above the identification label.

e. The size of the label and method of attachment shall be as shown in operation 17b of table II.

f. The label shall show fastness to laundering.

g. The contents of the size label shall be as follows for the specific size:

X-Small	Small	Medium
Chest: Up to 33 in. Stock No. NATO Size: 7080/7484	Chest: From 33 to 37 in. Stock No. NATO Size: 7080/8494	Chest: From 37 to 41 in. Stock No. NATO Size: 7080/9404

Large	X-Large
Chest: From 41 to 45 in. Stock No. NATO Size: 7080/0414	Chest: From 45 in. and upward Stock No. NATO Size: 7080/1424

h. Format and contents of the instruction label shall be as follows:

AT ALL TIMES KEEP THE "USE AND CARE" BOOKLET IN THIS POCKET

Important Instructions

1. This body armor system consists of coat with a removable collar and groin armor, trouser with spats, chest with face shield and helmet with a bonnet. You must wear the complete system in order to protect yourself against high velocity fragments.

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2. Be sure that all the straps, quick releases, buckles, chin strap, suspension system, and other components are properly adjusted.

3. Be sure that the face shield is clear of smudges or anything that would impede vision.

4. Keep the face shield covered and save it when not in use.

5. Keep the complete Body Armor System well covered from outer dust, dirt and moisture when not in use.

Cleaning Instructions:

1. Close fastener and outer cover surface and remove loose dirt using a cloth or soft brush. NEVER USE A STIFF BRISTLE BRUSH.

2. Wet any part of the suit that needs cleaning, using warm, not hot water.

3. Apply toilet soap or detergent to the soiled areas and scrub with a soft brush. Badly soiled areas may be scrubbed with GI soap. Scrub only long enough to remove soil.

4. Grease and oil stains may be pre-spotted with a detergent mixture and scrubbed with a soft brush. If stubborn stain persists, repeat the above procedure.

5. Rinse the suit, bonnet and groin protector with warm water until suds are completely gone.

6. Hang the coat and trouser on separate hangers to dry, away from heat or open flame.

7. Slightly wipe the area of the face shield with soft tissue or an absorbent cotton fabric slightly dampened with commercial type lens cleaning compound.

TAKE CARE OF YOUR:

a. Coat and Trouser

After washing, rinse the suit with warm water until the suds are completely gone. Hang the coat and trouser to dry, away from heat or open flame by using a stick or pole through the arms of the coat and trouser suspenders. Repair small rips or tears on any area of coat or trouser by using a needle and Nomex Thread. Turn in either coat or trouser, if you can't repair because of open seams, large rips, defective Quick-release, or other badly damaged parts of the suits, including fastener tape and webbing.

b. Helmet

Use helmet "Use and Care Booklet."

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c. Bonnet

Use same technique described in (a) above.

d. Chest Plate/Face Shield

The chest plate or the face shield may be replaced separately if either one is damaged. The face shield should be replaced if scratched or stained so that it impedes vision, or if cracked or damaged. The chest plate should be replaced if cracked and/or damaged.

3.3.11 Size label for ballistic filler components. Each ballistic filler component of the ballistic filler shall have a stamped label showing the size. The label shall conform to type IV, class 9 of DDD-L-20. The fastness of the class 9 label shall be as specified for class 5 labels. The labels shall be printed in bold letters 1/4-inch in height and located in the approximate center of each ballistic filler or component on the outer surface.

3.4 Design. The coat consists of an aramid oxford cloth inner and outer shell with ballistic fillers. The coat has a removable protective collar, a front pocket for inserting chest protector plate, and attaching groin protector. The left shoulder and the left side is gripped with a quick release hook and loop fastener closure to enable the Ordnance Disposal Specialist to don and doff the coat quickly. The back of the coat armholes is unattached to the coat sleeves for greater forward arm movement.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way. The patterns provide 3/8 inch seam allowance.

3.5.1 Pattern parts. The components of the coat shall be cut from the materials specified in table I in accordance with pattern parts indicated.

TABLE I. List of pattern parts

Material	Nomenclature	Parts
Ballistic cloth, aramid	Back insert #1	4
	Back insert #2	4
	Back insert #3	4
	Back insert #4	4
	Front insert #1	4
	Front insert #2	4
	Front insert #3	4
	Front insert #4	4
	Sleeve insert #1	8
	Sleeve insert #2	8

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TABLE I. List of pattern parts (cont'd)

Material	Nomenclature	Parts
Ballistic cloth, aramid (cont'd)	Sleeve insert #3	8
	Sleeve insert #4	8
	Groin protector insert	16
	Neck stand insert	8
	Collarstand insert	8
	Collar insert	16
	Elbow protective shield insert	32
	Back shoulder protective shield insert	32
Oxford, aramid	Back outer ply	1
	Back inner ply	1
	Front outer ply	1
	Front inner ply	1
	Sleeve outer ply	2
	Sleeve inner ply	2
	Plate pocket	2
	Tool pocket	1
	Groin protective outer	1
	Groin protective inner	1
	Neck stand	2
	Collarstand	1
	Collar	2
	Front shoulder extension	2
	Back shoulder extension	2
	Side closure flap	2
	Groin holder flap	1
	Collarstand extension	2
	Collar extension	1
	Tool pocket tab	2
	Tool pocket facing	1
	Elbow shield inner and outer	4
	Back shoulder shield inner and outer	4
	Sleeve elbow gusset inner and outer	4
	Sleeve tab	4

3.6 Construction. The construction shall conform in all respects to the requirements specified in table II and herein.

3.6.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances, shall be maintained with seams

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sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. The height of overedge stitching shall be 3/16 to 1/4 inch. Unless otherwise specified, edge stitching shall be 1/4 to 5/16 inches from edge.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 502, 503, 504 and 505 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching. All repairs shall be in accordance with 3.6.1.1.1a and b.

3.6.1.2.1 Optional type 101 holding stitching. Any additional holding stitching to facilitate manufacturing is permissible provided that during ballistic filler assembly the stay stitching is 1/4 ± 1/16 inch from edge of ballistic filler.

3.6.2 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met, and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

3.6.3 Thread ends. All thread ends that are visible on the finished coat shall be trimmed to a length of not more than 1/4 inch.

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3.6.4 Lubrication of thread. Except for fabrication of the ballistic filler components (see 6.5), there shall be no lubrication of the thread by any means prior to or during sewing (see 4.4.1.1).

3.6.5 Location marks. Marks for showing location of component parts on the outer shells may be done with marking ink or by use of drill holes providing they are located such that the marks or hole are completely and permanently covered by another component part.

3.6.6 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the coat.

3.6.7 Fusing of ends of nylon webbing and tape. All ends of nylon webbing and tape shall be fused. The apparatus used to fuse the webbing and tape ends shall be capable of providing sufficient heat to provide a smooth edge and with the cut ends of the webbing and tape yarns all fused together. Fusing of the webbing and tape ends shall be accomplished prior to being assembled for stitching.

3.7 Manufacturing operations requirements. The coat shall be manufactured in accordance with operations specified in table II. The contractor is not required to follow the exact sequence of operations.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the coat in strict accordance with patterns furnished which show size, warp directional lines. placement of components and notches for proper assembling of parts. b. The fabric components visible on the exterior of the body armor shall not be drilled for positioning any part except where a component part, when attached will completely and permanently cover the drill hole. The use of drilled holes on the ballistic material is prohibited. c. Each webbing or fastener tape component shall be cut in the required length as indicated on patterns and all nylon webbing ends shall be fused.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Assemble filler components. The filler components shall consist of 16 plies, in sized groups of four plies each. a. Evenly align edges of plies within a group and overedge the periphery with a 3/8 to 1/2 inch wide stitch gauge.	502, 503, 504, or 505	SSa-1	8-14	T-50	T-50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Assemble filler components. (cont'd)						
	b. Position filler groups one on top of another from large to small, evenly aligned and stitch around periphery 1/2 to 3/4 inch from edge, catching all plies.	301	SSa-1	8-10	T-50	T-50	
4.	Assemble filler.						
	a. Butt underarm seam together and baste in place (holding stitch).	200			T-50	T-50	
	b. To the large ply side of sleeve, position 1-1/2 inch webbing over basted seam and stitch with two rows of stitching on each side of webbing.	301	LSp-4	8-10	T-50	T-50	
	c. With smaller ply of coat body up, butt right shoulder seam. Positioning 1-1/2 inch webbing over seam, stitch with two rows of stitching on each side of webbing.	301	LSp-4	8-10	T-50	T-50	
	d. With all webbing towards the inside of filler, butt sleeves to coat body (per marks on patterns) and baste in place (holding stitch).	200			T-50	T-50	
	e. To the inside of armhole seam, position 1-1/2 inch webbing over basted seam and stitch with two rows of stitching on each side of webbing.	301	LSp-4	8-10	T-50	T-50	
	f. Butt right side seam and position 1-1/2 inch webbing to the inside of coat over butted seam, stitch with two rows of stitching on each side of webbing.	301	LSp-4	8-10	T-50	T-50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Assemble groin, elbow, and shoulder protectors.						
	a. With right sides together stitch elbow protector darts closed tacking ends.	301	SSa-1	8-10	T-35	T-35	
	b. Overedge stitch periphery of all groin, elbow, and shoulder protector shell pieces.	502, 503, 504 or 505		8-14	T-35	T-35	
	c. Position a 3 inch strip of 1-1/2 inch elastic webbing to the inner ply of each elbow protector towards point, turn the end under and box stitch in place.	301	Box-x	8-10	T-35	T-35	
	d. With right sides together stitch groin protector shell leaving top open to insert filler and tack ends. Repeat for elbow protectors and shoulder protectors.	301	LSq-1	8-10	T-35	T-35	
	e. Turn protectors right side out and turning seams to one side, raise stitch 1/32 to 1/16 inch from edge.	301	LSq-2	8-10	T-35	T-35	
	f. Insert fillers (snug-fitting at bottom and sides). Stay stitch 5/8 to 3/4 inch from edge around sides and bottom of shoulder protectors and groin protector, and per marks on patterns for elbow protectors.	301	SSa-1	8-10	T-35	T-35	
	g. Close protector shells by stitching close to fillers, producing snug fit.	301	SSa-1	8-10	T-35	T-35	
	h. With sufficient lengths of hook and loop fastener tape placed back to back, aligned, and completely containing groin protector top seam, stitch around all four sides of fastener tapes.	301	SSa-1	8-10	T-35	T-35	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Assemble groin, elbow, and shoulder protectors. (cont'd)						
	i. Position approximately 12 inches of 1-1/2 inch elastic webbing to the loop fastener tape side of the groin protector per marks on patterns, turn the end under and stitch with a box-x.	301	SSb-1	8-10	T-35	T-35	
	j. Turn the unsewn end of groin protector elastic webbing under to the same side as the opposite end and position a 5 inch strip of hook fastener tape and stitch all four sides 1/16 to 1/8 inch from edge.	301	SSb-1	8-10	T-35	T-35	
	k. Fold a 9 to 10 inch strip of 1-1/2 inch elastic webbing in half, butting ends. Position a 2-1/4 to 2-1/2 inch strip of 1-1/2 inch hook fastener tape to one end covering butted seam and stitch on all four sides 1/16 to 1/8 inch from edge.	301	SSa-1	8-10	T-35	T-35	
	l. Position a 10-1/2 to 11 inch strip of elastic webbing to shoulder protector per marks on pattern, turn end under 1/4 to 1/2 inch and stitch with a box-x per marks on pattern.	301	SSe-1	8-10	T-35	T-35	
	m. Turn raw end of left elastic webbing under 1/2 to 3/4 inch and position 3 inches of loop fastener tape to the under side. Stitch on all four sides 1/16 to 1/8 inch from edge.	301		8-10	T-35	T-35	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Assemble collar quick-release strap.						
	a. Position the longest piece of 1 inch wide quick-release webbing with one end wrapped around a 3 inch toggle and stitch with two individual rows of stitching or a box stitch.	301	SSa-2	8-10	T 35	T 35	
	b. Stitch all four sides of 2 to 2-1/2 inches of 1 inch hook fastener tape to longest quick-release webbing as indicated on patterns.	301	SSa-1	8-10	T 35	T 35	
	c. Turning end of short quick-release webbing under 1/4 to 3/8 inch, stitch to longer quick-release webbing on the opposite side as hook fastener tape and as indicated on patterns, using a 1 inch box x stitch.	301		8-10	T-35	T 35	
	d. Fold collar extension in half with right sides together and stitch both sides.	301	SSa-1	8-10	T-35	T 35	
	e. Turn collar extension right side out, pushing out corners, and edge stitch 1/16 to 1/8 inch from sides and end.	301	SSe-2	8-10	T-35	T-35	
	f. Butt the sides of two pieces of 1-1/2 inch hook fastener tape sufficient in length to reach side to side and stitch on all four sides of both pieces.	301	SSa-1	8-10	T-35	T 35	
	g. Turn end of short quick-release webbing under 1/4 to 3/8 inch and in the same direction as opposite end and stitch to collar extension, centered and on opposite side from hook fastener tape using a 3/4 to 1 inch box stitch.	301	SSb-1	8-10	T-35	T-35	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Assemble collar quick-release strap. (cont'd)						
	h. With right sides together, stitch collarstand extension on sides and point.	301	SSa-1	8-10	T-35	T-35	
	i. Turn collarstand extension right side out, pushing out points, and edge stitch 1/16 to 1/8 inch around sides and point.	301	SSe-2	8-10	T-35	T-35	
	j. Position 2 to 2-1/4 inches of 1-1/2 inch hook fastener tape towards point of collarstand extension and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
	k. Turn end of long quick-release webbing towards hook fastener tape 1/4 to 3/8 inch and stitch to collarstand extension point on opposite side from hook fastener tape, using a 1/4 to 1/2 inch box stitch.	301	SSb-1	8-10	T-35	T-35	
7.	Assemble collarstand.						
	a. Overedge periphery of all collarstand shell pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. Turn one end of the collarstand under 3/8 to 1/2 inch and stitch 1/32 to 1/16 inch from edge.	301	EFa-1	8-10	T-35	T-35	
	c. Position 3/4 to 1 inch pieces of 1-1/2 inch hook and loop fastener tapes over collarstand hem and stitch on all four sides of both.	301	SSa-1	8-10	T-35	T-35	
	d. Fold collarstand in half lengthwise with right sides together. Position collarstand extension between insides of opposite end and stitch end closed tacking securely.	301	SSj	8-10	T-35	T-35	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Assemble collarstand. (cont'd)						
	e. Turn collarstand right side out pushing out corners. Fold collarstand in half lengthwise and edge stitch along fold and across extension end, tacking securely.	301	SSe-1	8-10	T-35	T-35	
	f. Position 1-1/2 inch loop fastener tape of sufficient length to cover the collarstand to the outside of collarstand and on the same side as collarstand extension hook fastener tape, and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
	g. Position 2 to 2-1/2 inches of 1-1/2 inch loop fastener tape at open end of collarstand on the outside and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
	h. Insert collarstand filler, snug-fitting along outside and extension end. Stitch the length of stand as close to filler as possible, tacking both ends.	301	SSa-1	8-10	T-35	T-35	
8.	Assemble collar.						
	a. Overedge periphery of all collar shell pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. With right sides together join outside length of collar.	301	SSa-1	8-10	T-35	T-35	
	c. Turn right side out and turning the seam to one side, raise stitch 1/32 to 1/16 inch from edge.	301	LSq-2	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Assemble collar. (cont'd)						
	d. Position extension on the right side and towards the right end of collar. With fastener tape down, turn under 1/4 to 3/8 inch of extension and stitch with a box formation as indicated on patterns. NOTE: Check for twists in quick-release strap before stitching.	301	SSb-1	8-10	T-35	T-35	
	e. Fold the collar in half lengthwise, right sides together and stitch the extension end of collar.	301	SSa-1	8-10	T-35	T-35	
	f. Lay the open end of collar flat, wrong side up and turn the end in 1/8 to 3/8 inch. Position alternate strip of 1 inch hook and loop fastener tape over the folded end and stitch on all four sides.	301	SSa-2	8-10	T-35	T-35	
	g. To the outside, position two strips of 1-1/2 inch loop fastener tape butted and as indicated on patterns and stitch on all four sides of both.	301	SSa-1	8-10	T-35	T-35	
	h. Turn collar wrong side out, place collarstand along length between insides and stitch all plies.	301	SSj-1	8-10	T-35	T-35	
	i. Turn collar right side out and insert collar filler.						
9.	Assemble neck stand.						
	a. Overedge periphery of all neck stand pieces.	502, 503, 504, or 505		8-14	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Assemble neck stand. (cont'd)						
	b. Position six 3-1/2 by 3-3/4 inch pieces of 1-1/2 inch hook fastener tape centered and evenly spaced on right side of neck stand and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
	c. With right sides together stitch ends and outer length.	301	SSa-1	8-10	T-35	T-35	
	d. Turn right side out (pushing out points) and edge stitch 1/32 to 1/16 inch from edge.	301	SSe-2	8-10	T-35	T-35	
	e. Insert neck stand filler and stitch as close to filler as possible to insure a snug fit.	301	SSa-1	8-10	T-35	T-35	
10.	Assemble groin protector holding flap.						
	a. Overedge periphery of all groin holder flap pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. Fold flap in half lengthwise and stitch ends.	301	SSa-1	8-10	T-35	T-35	
	c. Turn flap right side out, pushing out corners.						
	d. Position a strip of 1 inch hook fastener tape the length of the flap and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
11.	Assemble tool pocket.						
	a. Overedge periphery of tool pocket.	502, 503, 504, or 505		8-14	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Assemble tool pocket. (cont'd)						
	b. Per marks on pattern fold side and bottom extensions to the right side and edge stitch along the fold 1/32 to 1/16 inch from edge.	301	SSb-2	8-10	T-35	T-35	
	c. With right sides together stitch corner seam.	301	SSa-1	8-10	T-35	T-35	
	d. Turn right side out and raise stitch 1/32 to 1/16 inch from edge of corner seam.	301	LSq-2	8-10	T-35	T-35	
	e. Per second marks on pattern fold side and bottom to the inside and edge stitch along the fold.	301	SSb-2	8-10	T-35	T-35	
	f. With right sides together stitch pocket opening facing to pocket opening.	301	SSa-1	8-10	T-35	T-35	
	g. Turn to the wrong side, turn raw edge under and stitch with a 5/8 inch gauge double row of stitching 3/8 to 1/4 inch from edge.	301	EFb-1	8-10	T-35	T-35	
	h. Per marks on pattern stitch 2-1/4 to 2-1/2 inches of 2 inch loop fastener tape to outside of pocket opening on all four sides 1/8 to 1/4 inch from edge.	301	SSa-1	8-10	T-35	T-35	
12.	Assemble tool pocket and sleeve tabs.						
	a. Overedge periphery of tab pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. With right sides together stitch sides and point of each tab.	301	SSa-1	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Assemble tool pocket tab and sleeve tabs. (cont'd)						
	c. Turn right side out (pushing out point) and top stitch 3/8 to 1/4 inch from edge of each tab.	301	SSe-1	8-10	T-35	T-35	
	d. Position 3 to 3-1/2 inches of hook fastener tape to pocket tabs and loop fastener tape to sleeve tabs, and stitch on all four sides 1/16 to 1/8 inch.	301	SSa-1	8-10	T-35	T-35	
13.	Assemble plate pocket (with adjusting straps).						
	a. Overedge periphery of each plate pocket piece.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. With right sides together stitch around sides and top.	301	SSa-1	8-10	T-35	T-35	
	c. Turn right side out (pushing out points) and top stitch 3/8 to 1/4 inch from edge.	301	SSe-1	8-10	T-35	T-35	
	d. Thread the 2-1/2 to 3 inch adjusting straps thru the buckles to the half point, fold over and stitch with three overlapping rows of stitching or by using a 1/2 inch bartack.	301 or bartack	SSb-1	8-10 28 per bartack	T-35 T-35	T-35 T-35	
	e. Thread the 24 to 25 inch left adjusting strap and the 18 to 19 inch right adjusting strap thru the slider end of buckles, lay flat with buckle right side up, turn the end up 1/4 to 1/2 inch and position 7 inches of loop fastener tape along the left strap and 7 inches of loop fastener tape along the right strap and stitch on all four sides.	301	SSa-2	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Assemble plate pocket (with adjusting straps). (cont'd)						
	f. Position the raw end of left and right adjusting straps with sides to each plate pocket top left and right corner, as marked on patterns and stitch with a box-X pattern.	301	SSa-1	8-10	T-35	T-35	
14.	Assemble back shoulder extension.						
	a. Overedge stitch periphery of extension pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. With right sides together, stitch sides and point together.	301	SSa-1	8-10	T-35	T-35	
	c. Turn right side out (pushing out points) and top stitch 3/8 to 1/4 inch from edge.	301	SSe-1	8-10	T-35	T-35	
	d. Butt two pieces of 2 inch hook fastener tape the width of the extension and stitch on all four sides of each.	301	SSa-1	8-10	T-35	T-35	
15.	Assemble front shoulder extension (with quick-release strap).						
	a. Overedge stitch periphery of extension pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. With right sides together, stitch sides and point together.	301	SSa-1	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Assemble front shoulder extension (with quick-release strap). (cont'd)						
	c. Turn right sides out (pushing out points) and top stitch 3/8 to 1/4 inch from edge.	301	SSe-1	8-10	T-35	T-35	
	d. Butt two pieces of 2 inch hook fastener tape the width of the extension and stitch on all four sides.	301	SSa-1	8-10	T 35	T 35	
	e. Fold cotton webbing in half, position as marked on patterns to the opposite side from fastener tape and stitch using a box-x stitch.	301	SSa-1	8-10	T-35	T-35	
	f. Position long piece of webbing at the point of shoulder extension, turn under 1/2 inch and stitch using a box-x stitch.	301	SSb-1	8-10	T-35	T-35	
	g. Laying the webbing flat, stitch a 2 inch piece of 1 inch hook fastener tape centered on webbing and half way down the length and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
	h. Turning webbing over, position 2 inches of 1 inch loop fastener tape, starting in the middle of the hook fastener tape stitching, towards raw end and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	
16.	Assemble side closure flap.						
	a. Overedge stitch periphery of flap pieces.	502, 503, 504, or 505		8-14	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Assemble side closure flap. (cont'd)						
	b. With right sides together stitch sides and point together.	301	SSa-1	8-10	T-35	T-35	
	c. Turn right side out (pushing out point) and top stitch 3/8 to 1/4 inch from edge.	301	SSe-1	8-10	T-35	T-35	
	d. Butt and/or overlap five 6 inch strips of 2 inch hook fastener tape across the width of flap and one 9 inch strip lengthwise at the bottom of others, per marks on patterns, and stitch on all four sides of each.	301	SSa-1	8-10	T-35	T-35	
	e. Position the short piece of quick-release webbing at the point end of flap, on opposite side of fastener tape. Turn the end under 1/4 to 3/8 inch and box stitch reinforcing the point side of stitching.	301	SSb-1	8-10	T-35	T-35	
	f. Lay webbing flat (with fastener tape on flap up) and turn raw end under 3/8 to 1/2 inch. Position the 3 inch strip of webbing encompassing the long toggle, centered on the end and box-x stitch reinforcing the toggle side.	301	SSa-1	8-10	T-35	T-35	
	g. Position raw end of long quick-release strap (attached to shoulder extension) to the short strap on opposite side of toggle webbing and box stitch in place.	301	SSa-1	8-10	T-35	T-35	
	h. Position a 5 inch strip of 2 inch hook fastener tape at the end of joined straps on the opposite side of toggle webbing, and stitch on all four sides.	301	SSa-1	8-10	T-35	T-35	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Assemble coat body inner shell.						
	a. Overedge periphery of both front and back piece.	502, 503, 504, or 505		8-14	T 35	T 35	
	b. The combined instruction, size, and identification label (see 3.3.9) shall serve as a pocket for the instruction booklet and shall be fabricated and attached to the outer surface of the back inner shell. The label shall be stitched at the center of inner shell approximately 5 inches down from neck seam. The row of horizontal stitching required to form the bottom of the pocket shall not pass through the printing.	301	SSa-1	8-10	T-35	T 35	
	c. Position two 5 inch strips of butted 2 inch loop fastener tape on inside left front shoulder per marks on patterns, and stitch on all four sides of both.	301	SSa-1	8-10	T-35	T-35	
	d. With right sides together stitch front to back at right shoulder and right side seam.	301	SSa-1	8-10	T-35	T 35	
	e. Turn right side out, with seams towards back and raise stitch 1/32 to 1/16 inch from edge.	301		8-10	T-35	T-35	
18.	Assemble coat body outer shell.						
	a. Overedge periphery of front and back pieces.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. Position five 7 inch strips of butted 2 inch loop fastener tape at outer back left side per marks on patterns and stitch on all four side of each.	301	SSa-1	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Assemble coat body outer shell. (cont'd)						
	c. Position two 5 inch strips of butted 2 inch loop fastener tape at outer left back shoulder as marked on patterns, and stitch on all four sides of each.	301	SSa-1	8-10	T-35	T-35	
	d. Position a 4-3/4 to 5 inch strip of 1 inch loop fastener tape to outer lower front per marks on pattern and stitch on all four sides 1/16 to 1/8 inch from edge.	301	SSa-1	8-10	T-35	T-35	
	e. Position a 4-3/4 to 5 inch strip of 2 inch loop fastener tape to the outer back centered along bottom and stitch on all four sides 1/16 to 1/8 inch from edges.	301	SSa-1	8-10	T-35	T-35	
	f. Turn raw end of right shoulder shield elastic webbing under 1/2 to 3/4 inch and position 3 inches of hook fastener tape to the under side and stitch on all four sides 1/16 to 1/8 inch from edge.	301	LSs-1	8-10	T-35	T-35	
	g. Position side closure flap with fastener tape up and laying towards center, and per marks on patterns, and stitch across flap.	301	SSa-1	8-10	T-35	T-35	
	h. Fold closure flap over towards side and top stitch 1/4 to 3/8 inch from fold, tacking both ends.	301	LSs-1	8-10	T-35	T-35	
	i. Position tool pocket to outer front per marks on patterns, fold under seam allowance on side and bottom and edge stitch 1/32 to 1/16 inch from edge.	301	SSb-1	8-10	T-35	T-35	
	j. Position pocket tab to outer front per marks on patterns with fastener tape up and tab pointed away from pocket, and stitch across tab.	301	SSa-1	8-10	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Assemble coat body outer shell. (cont'd)						
	k. Fold tab over towards pocket and top stitch 1/4 to 3/8 inch from fold, tacking both ends.	301	SSb-1	8-10	T-35	T-35	
	l. Position the bottom of plate pocket per marks on pattern with the top of pocket towards the bottom of coat and webbing facing down. Stitch in place catching the top of the tool pocket.	301	SSa-1	8-10	T-35	T-35	
	m. Fold plate pocket towards the top of the coat and top stitch 1/4 to 3/8 inch from edge along half side seams and across the bottom, tacking ends.	301	SSb-1	8-10	T-35	T-35	
	n. Tack plate pocket adjusting straps to shoulders.	301	SSa-1	8-10	T-35	T-35	
	o. With right sides together, join front to back at right shoulder and right side seam.	301	SSa-1	8-10	T-35	T-35	
	p. Turn right side out with seams toward front and raise stitch 1/8 to 3/32 inch from edge.	301		8-10	T-35	T-35	
	q. Position neck stand with hook fastener tape towards right side of coat and stitch to neck line, tacking ends.	301		8-10	T-35	T-35	
	r. Position left back shoulder extension to back left shoulder with hook fastener tape facing loop fastener tape, and seam.	301	SSa-1	8-10	T-35	T-35	
	s. Position left front shoulder extension to front left shoulder with hook fastener tape facing loop fastener tape, and seam.	301	SSa-1	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Assemble sleeves.						
	a. Overedge stitch periphery of inner and outer sleeve plies.	502, 503, 504, or 505		8-14	T-35	T-35	
	b. With right sides together stitch elbow gussets and sleeve gussets to inner and outer plies.	301	SSa-1	8-10	T-35	T-35	
	c. Turn gusset seam allowances towards gusset at the elbow and towards sleeve at wrist and edge stitch 1/32 to 1/16 inch from edges.	301	SSe-1	8-10	T-35	T-35	
	d. Position shoulder protectors and elbow protectors to outer sleeves per marks on patterns with protectors pointing away from shoulders and with webbing down from shoulder protectors and up for elbow protectors, and stitch in place tacking ends.	301	SSa-1	8-10	T-35	T-35	
	e. Turn shoulder protectors and elbow protectors toward the shoulder and raise stitch 5/8 to 3/4 inch from edge.	301	SSe-1	8-10	T-35	T-35	
	f. Position loose end of elbow protector webbing to sleeve per marks on patterns, turn end under and stitch with box-x.	301	Box-x	8-10	T-35	T-35	
	g. Position a 2-1/2 to 3 inch strip of 1-1/2 inch hook fastener tape to the outer sleeves per marks on patterns at wrist and stitch on all four sides 1/16 to 1/8 inch from edge.	301	SSa-1	8-10	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Assemble sleeves. (cont'd)						
	h. Position sleeve tabs with point away from gussets and fastener tape up per marks on patterns and stitch on all four sides 1/16 to 1/8 inch from edge.	301	SSa-1	8-10	T-35	T-35	
	i. Turn sleeve tabs towards gussets and raise stitch 1/4 to 3/8 inch from folded edge.	301	SSe-1	8-10	T-35	T-35	
	j. With sleeve folded right sides together join along seam.	301	SSa-1	8-10	T-35	T-35	
	k. Turn right side out and folding seam away from dart raise stitch 1/16 to 1/8 inch from seam.	301		8-10	T-35	T-35	
	l. Insert inner sleeve into outer sleeve and turning in seam at bottom edge stitch 1/32 to 1/16 inch from edge.	301	SSc-1	8-10	T-35	T-35	
20.	Assemble coat.						
	a. With right sides together join inner coat to outer coat, leaving side and bottom open for turning. Leave armholes open.	301	SSa-1	8-10	T-35	T-35	
	b. Insert filler into body shell and sleeves.						
	c. Per marks on pattern stitch sleeves to body.	301	SSa-1	8-10	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
20.	Assemble coat. (cont'd)						
	d. Form armhole opening by stitching inner and outer sleeve together and inner and outer coat together per marks on patterns. Catch left shoulder extension elastic webbing in seam when closing left sleeve opening.	301	SSc-1	8-10	T-35	T-35	
	e. With seam towards sleeve raise stitch around armhole and sleeve 1/32 to 1/16 inch from edge.	301		8-10	T-35	T-35	
	f. Bartack in all stress points (18) per marks on patterns.	Bartack		21 per bartack	T-35	T-35	
	g. Close the side and bottom by turning the seams in and edge stitch 1/16 to 1/8 inch from edge.	301	SSc-1	8-10	T-35	T-35	
	h. Stitch through all plies for holding stitch per marks on patterns (7), tacking both ends.	301		8-10	T-35	T-35	
21.	Clean coat.						
	a. Trim and remove all thread ends.						
	b. Remove all spots, stains, and size markings.						

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3.8 Finished dimensions. The finished dimensions for the body armor coat shall conform to table III.

TABLE III. Finished dimensions (inches)

Sizes	Extra Small	Small	Medium	Large	Extra Large	Tolerance
Across chest, below armholes from side seam to side seam	21	23	25	27	29	± 1/2
Across back, below armholes from side seam to side seam	21	23	25	27	29	± 1/2
Coat back length 1/	27	27-1/2	28	28-1/2	29	± 1/2
Sleeve (inseam) length	19-5/8	19-3/4	19-7/8	20	20-1/8	± 1/4
Groin 2/ and	12	13	14	15	16	± 1/4
Groin 3/	10	10-1/2	11	11-1/2	12	± 1/8

1/ Measure from neck seam to bottom of coat.

2/ Measure across top edge of groin.

3/ Measure across center, from top edge of hook pile fastener tape to bottom edge of groin.

3.9 Workmanship. The finished body armor coat shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Utmost care shall be taken during component and subassembly fabrication to ensure quality workmanship and safety of the service person using the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the thread lubrication restriction conforms to the requirement specified in 3.6.4.

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4.4.2 In-process inspection.

4.4.2.1 In-process examination of cut parts, fillers, and patterns. All cut parts for the shell and filler assemblies shall be examined during the cutting process to determine that parts are cut properly with respect to size and material directional requirements, and that parts containing defects such as a hole, cut, smash, float, loose slub, hard crease or wrinkle are removed from production. All ballistic fillers shall be examined during assembly of the individual groups to ensure that they contain the correct amount of plies. In addition to the above, inspection shall be made of the working patterns to ensure that they conform to the Government patterns in all respects. Whenever nonconformance is noted, correction shall be made to the item or items affected. Parts which cannot be corrected shall be removed from production.

4.4.2.2 In-process size examination for filler groups. The ballistic filler groups after edge stitching shall be examined for the defects listed in table IV. The lot size shall be expressed in terms of left, right, or center filler groups, as applicable. The sample unit shall be one left, right, or center filler group, as applicable. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total (major and minor combined) defects.

TABLE IV. In-process size defects

Examine	Defect	Classification Major	Minor
Size of individual ballistic filler	a. Smaller than cutting pattern 3/16 inch to 3/8 inch at any point around the periphery 1/ 2/		201
	More than 3/8 inch at any point around the periphery 1/ 2/	101	
	1/8 inch up to 3/16 inch around entire periphery 2/		202
	More than 3/16 inch around entire periphery 2/	102	
	b. Larger than cutting pattern by 3/8 inch or more at any point 1/ 2/		203

1/ To be scored when condition exists for a length of more than 2 inches or if condition exists in several areas with an accumulated distance of 4 inches or more.

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2/ The individual component, as applicable, shall be examined with the applicable cutting pattern centered on the filler component.

4.4.3 End item visual examination. The end items shall be examined for the defects specified in table V. The lot size shall be expressed in units of coats. The sample unit shall be one completed coat with groin protector and collar. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15 for total (major and minor combined) defects.

TABLE V. End item visual defects

Examine	Defect	Classification Major	Minor
Cloth	Any hole, (except drill hole) cut, or tear	101	
	Any exposed drill hole		201
	Any abrasion marks, broken or missing yarns, or multiple floats	102	
	Any mend, darn, or patch	103	
	Parts not cut correctly with respect to material lines	104	
	Needle chews	105	
Tape	Any hole, cut, or frayed edge	106	
	Needle chews:		
	(1) Up to 1/8 inch in length		202
	(2) 1/8 inch or more in length	107	
	Not continuous, i.e., spliced	108	
Seams and stitchings:			
Open seams	Up to and including 1/2 inch		203
	More than 1/2 inch	109	
	NOTE: A seam shall be classified as an open seam when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur		

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitchings: (cont'd)			
Raw edges (except where required or otherwise classified herein)	More than 1/2 inch when securely caught in stitching		204
NOTE:	Raw edges not securely caught in stitching shall be classified as open seams		
Seam and stitch type	Wrong seam or stitch type	110	
	Seam pleated or badly puckered clearly noticeable		205
Stitch tension	Tension loose, resulting in loose bobbin or top thread		206
	Tension tight, resulting in puckering of material		207
Stitches per inch	One to two stitches less than minimum specified 1/		208
	Three or more stitches less than minimum specified 1/	111	
	One or more stitches in excess of maximum specified 1/		209
NOTE:	Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows:		
	With the minor defect classification - No defect		
	With the major defect classification - Minor defect		

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitchings: (cont'd)			
Thread break, or skipped stitches (on type 301 stitching)	Overstitched less than 1 inch in each direction beyond defective stitching area		210
	NOTE: On all types of stitching, thread breaks or two or more consecutive skipped stitches not overstitched shall be classified as open seam		
Stitching ends (on type 301 stitching)	Ends of stitching not secured as specified (except when caught in other stitching or turned under in a hem)		211
Runoffs	On joining or cl sing seams - use open seam classification		
	Edge or raise stitching, not resulting in any open seam, more than 1/2 inch in length (when resulting in an open seam use open seam classification)		212
Rows of stitching	Any row missing (unless otherwise classified herein)	112	
Assembly - general	Any component caught in any unrelated operations of stitching, affecting appearance or function (not otherwise classified herein)	113	
	Any component part omitted (not otherwise classified herein) or not attached as specified	114	
	Any row of stitching securing ballistic filler to inner shell omitted	115	

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Labels (size, identification and instruction)	Missing, incorrect, or illegible	116	
	Misplaced		213
	Color not as specified	117	
Cleanness	Any spot or stain clearly noticeable		214
	Thread ends not trimmed as specified		215
Color of components	Color of any component not as specified		216

1/ Defects to be scored only when condition exists for a length of 3 inches or more or in several areas with an accumulated distance of 5 inches or more.

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table III. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of coats. The sample unit shall be one coat with groin protector. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: not properly secured, incomplete sealing or closure of box flap, improper taping or tying, loose strapping or inadequate stapling Bulged or distorted container
Content	Number of items per container is more or less than required

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4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Each coat, with the collar removed and all fasteners closed, shall be laid out flat, front up. The collar shall be folded in half and placed lengthwise on the front of the coat. The groin protector shall be folded up and the sleeves folded diagonally across the front of the coat. The folded coat shall measure approximately 31 inches in length by 24 inches in width.

5.1.2 Commercial preservation. Coats shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. One coat, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style FTC, grade V2s of PPP-B-636. Coats shall be completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. The inside dimensions of the container shall approximate 38 inches in length, 15-1/2 inches in width, and 3-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. One coat, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FTC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Coats shall be completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. The approximate inside dimensions of the shipping container shall be as specified in 5.2.1. Each fiberboard shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Coats, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), coats packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The body armor is intended for use for protection against high velocity fragments associated with Unexploded Ordnance (UXO) and Improvised Explosive Devices (IED).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Size required (see 1.2).

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d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.2, 4.3, and 6.3).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Type and class of unit load required (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Cooling of needle. In order to keep the needle cool when overedging the ballistic fillers, it is recommended tha either the machines be equipped with a compressed air blower directed at the needle or by affixing a felt pad saturated with silicone oil in a position so that the thread will rub against it during sewing. The use of chromium plated needles have been found to be advantageous in that they do no generate as much heat or deflect as much during use as regular needles. Recommended maximum speed for the overedge machine is 1000 RPM.

6.6 Life support clothing and equipment. All Government administrative and surveillance and quality verification provisions applicable to LSC&E items shall be invoked for items purchased in accordance with this specification.

6.7 Subject tern (key word) listing.

Aramid
Ballistic protection
IED
Improvised explosive device
Life support clothing
LSC&E
Protective clothing
Unexploded ordnance
UXO

6.8 Changes from previous issue. Marginal notation are not used in this revision to identify changes with respect to the previous iuuse due to the extensiveness of the changes.

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Air Force - 99

Review activities:

Army - MD
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DLA - CT

Preparing activity:

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